

Lab 2

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Downloading the PCAP File

```
PS C:\Users\Stephanie\Desktop> ls

Directory: C:\Users\Stephanie\Desktop

Mode                LastWriteTime         Length Name
----                -
-a----             9/5/2024  11:16 AM        179298 wpa-Induction.pcap

PS C:\Users\Stephanie\Desktop> |
```

I made sure I downloaded the file to my desktop.

Viewing the EAPOL Handshake

87	5.649953	CiscoLinksys_82:b2:...	Apple_82:36:3a	EAPOL	181	Key (Message 1 of 4)
88	5.649964		CiscoLinksys_82:b2:...	802.11	38	Acknowledgement, Flags=.....C
89	5.650959	Apple_82:36:3a	CiscoLinksys_82:b2:...	EAPOL	181	Key (Message 2 of 4)
90	5.650970		Apple_82:36:3a	802.11	38	Acknowledgement, Flags=.....C
91	5.654947		CiscoLinksys_82:b2:...	802.11	38	Clear-to-send, Flags=.....C
92	5.655957	CiscoLinksys_82:b2:...	Apple_82:36:3a	EAPOL	239	Key (Message 3 of 4)
93	5.655968		CiscoLinksys_82:b2:...	802.11	38	Acknowledgement, Flags=.....C
94	5.655973	Apple_82:36:3a	CiscoLinksys_82:b2:...	EAPOL	159	Key (Message 4 of 4)

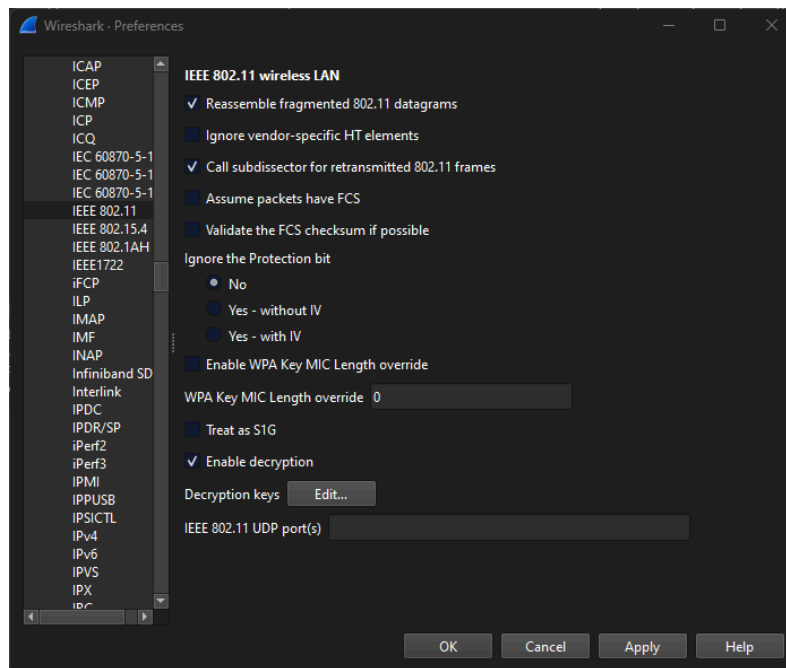
I start by opening the file in Wireshark. Then, I scroll down to find the frames with “EAPOL” protocol. These 4 packets are used to negotiate a private key.

Viewing Encrypted Traffic

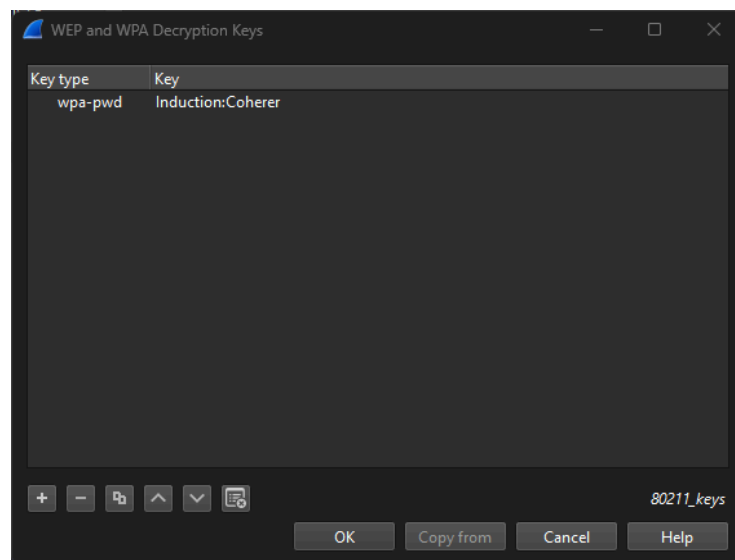
99	5.844024	Apple_82:36:3a	Broadcast	802.11	404	Data, SN=27, FN=0, Flags=.p....TC
Frame 99: 404 bytes on wire (3232 bits), 404 bytes captured (3232) on interface 0						
Radiotap Header v0, Length 24						
802.11 radio information						
IEEE 802.11 Data, Flags: .p....TC						
Data (344 bytes)						

Notice at line 99, the “Info” displayed by Wireshark is only “Data”, meaning Wireshark was unable to decrypt the contents of the frame.

Entering the WPA Key

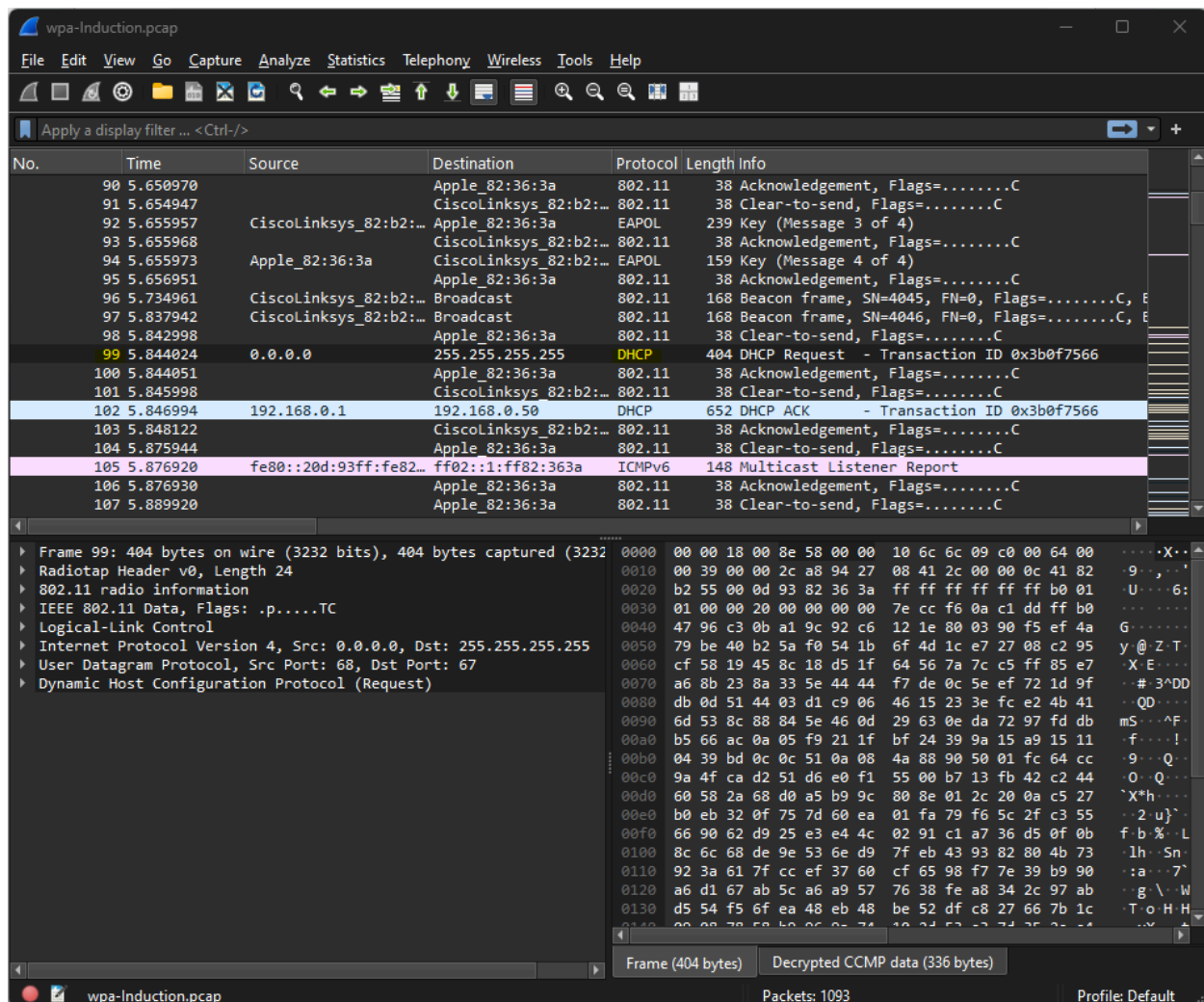


I navigate to "Edit" > "Preferences" > "Protocols" > "IEEE 802.11". Then, I click on "Edit..." next to "Decryption keys".



Then, I add a key of type "wpa-pwd" with a value of "Induction:Coherer", in which "Induction" is the key and "Coherer" is the SSID of the network.

Frame 99 Decrypted



After enabling decryption and adding a key, I view the file again. Notice this time, frame 99 displays “DHCP” protocol, revealing it contains a “DHCP” packet, confirming its contents have been decrypted.

Summary

In this lab, I got to work with Wireshark and view a provided “.pcap” file. Most notably, I learned that you can add keys to decrypt files, something that had never crossed my mind while using Wireshark in the past.